

TrES-2b

R-0130

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_170331 · created 2026-07-16T06:11:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457843.90947 +/- 0.00051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.04
Transit depth (Rp/Rs)^2	2.01 +/- 1.15 [%]
Orbital Inclination (inc)	82.81 +/- 2.53
Airmass coefficient 1 (a1)	1.0 +/- 0.015
Airmass coefficient 2 (a2)	0.0018 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#1 - [81, 132]
Optimal Aperture	4.49
Optimal Annulus	13.45
Transit Duration (day)	0.05 +/- 0.04

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.04 vs 0.1254 (deviation 0.39σ)
Duration fit vs published	0.05 d vs 0.0746 d (deviation 0.58σ)
Mid-time O–C	-1.3 min
Depth SNR	3.3
Residual scatter	1.47 %

Light curve

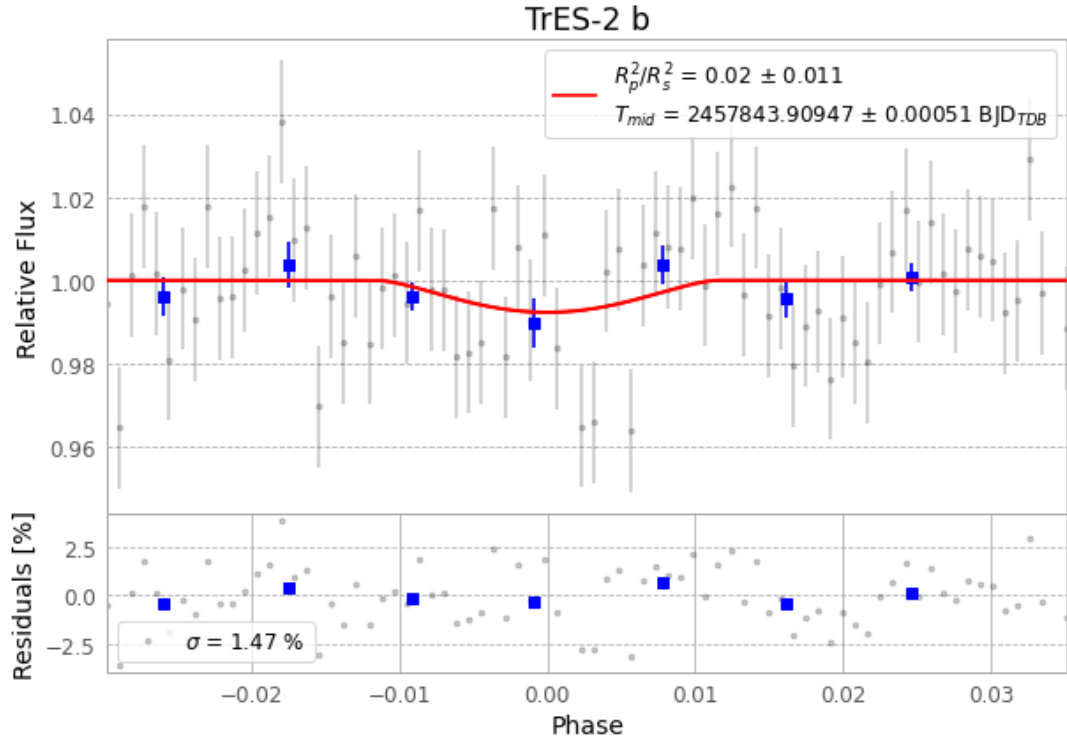


Figure 1 — FinalLightCurve_TrES-2 b_2017-03-31.png (SHA-256 1a04e0595d5070e9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [75, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c08afb880317d97...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

77 FITS frames (51 MB), TRES-2170331080312.FITS → TRES-2170331115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-170331.log (a09bfc2494df...), FinalLightCurve_TrES-2 b_2017-03-31.png (1a04e0595d50...), FinalLightCurve_TrES-2 b_2017-03-31.csv (3c56755b0ca0...)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:11Z.